Fats are absorbed by passive diffusion mainly in the proximal jejunum (Figure 2.22).

"Mixed micelle" formation: The fatty acids, mono and diglycerides and cholesterol, the enzymatic hydrolytic products of lipid digestion along with bile salts, together, form water-soluble molecular aggregates called as mixed "micelles of size 0.1 to 0.5 μm in diameter and are absorbed mainly from duodenum and jejunum. Micelles can also accommodate some fat soluble substances like carotene and vitamins.

- Micellar complexes get destabilized when they come near the brush border of enterocytes and liberates its contents.
- This is followed by passive diffusion of the digested fats across the luminal cell membrane.
- Only few fatty acids like oleic acid and linoleic acid enter the enterocyte by facilitated diffusion.
- In the endoplasmic reticulum of the enterocyte, triacylglycerols and cholesterol esters are re-synthesized
- These resynthesized lipids, together with small quantities of phospholipids and apolipoprotein B-48, form the chylomicrons.
- The chylomicrons cross the other side of the cell membrane and enter into the lymphatic vessels.
- The triacylglycerols of short chain fatty acids and few short chain free fatty acids of chain length (6–10 carbon atoms) can directly enter the portal circulation.

Bile salts of the "micelles" are not absorbed initially, while they are reabsorbed in the lower part of the small intestine and returned to the liver through the portal vein for re-secretion into the bile. Such a process is known as entero-hepatic circulation of bile salts.

Factors affecting absorption

- Short chain fatty acids enter the circulation at a faster rate than the long chain fatty acids.
- Certain plant sterols like stigmasterol and sitosterol inhibit cholesterol absorption. Drugs like statins and certain phytochemicals present in green tea decrease the absorption of fats.
- Bile salts enhance the digestion and absorption of fats. Presence of unsaturated free fatty acids enhances the absorption of cholesterol.
- Certain conditions like obstructive jaundice/ pancreatic disorders reduce the digestion and absorption of fats.

2.2.2.4 Digestion and absorption of nucleic acids

Digestion in mouth: No digestion of nucleic acid has been reported in the mouth.

Digestion in stomach: Nucleoproteins are denaturated and nucleotides are released by the acidic environment in the stomach. Recent reports suggest that the cleavage of nucleic acids is also facilitated by pepsin in the stomach.

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